

Pairing Proofs and Polynomial Ring Toolkit

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Abstract. In this work, we consider the setting where elliptic curve pairings need to be performed within another proving system. This appears in proof recursion where a proof verifies other proofs, or when pairing-based protocols like BLS (Boneh-Lynn-Shacham) signatures or KZG commitments are used within a proof. We show a construction that simultaneously reduces arithmetic circuit depth and communication complexity.

We present a *two-round public-coin interactive proof system* that consolidates the Miller loop iterations and final exponentiation into unified polynomial relations, verified as polynomial identities over $\mathbb{F}_q[x]$. This construction significantly reduces computational overhead across R1CS, Plonkish and AIR-based systems. We prove knowledge soundness $\delta_s \leq 2^{-245}$ for a three-pair optimal ate multi-pairing on BN254 curve.

«««< HEAD As our *second contribution* we present a generalised construction of our core technique, parameterised by an arbitrary modulus polynomial, designed for easy application across various protocols and schemes beyond pairings that utilize polynomial ring multiplications.

We demonstrate the practical efficacy of our construction across R1CS, Plonkish, and AIR arithmetisations on BN254 and BLS12-381, with implementations in gnark—the open-source zk-SNARK ecosystem—and in Garaga, the standard library for pairing-based ZK proof verification on Cairo, a leading AIR-based CPU architecture. ===== We present a full implementation in gnark [Cona]—the industry-standard framework for arithmetized circuit development—as a reusable polynomial ring toolkit in the `emulated` package. Benchmarked on a 3-pairing product check—the pairing workload of a Groth16 verification—our gnark implementation achieves **over 32% reduction in constraints, over 50% reduction in commitments**, and reductions in both compile-time and solver memory for both BN254 and BLS12-381 curves. These improvements hold across both SCS and R1CS constraint systems and across curves—BLS12-381 shows a 33% constraint and 54.5% commitment reduction—enabling practical pairing verification in recursive proof systems and resource-constrained environments. »»»> 485e970 (chore: abstract add bls 381)

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1 Introduction

Bilinear pairings on elliptic curves are a fundamental building block of modern cryptography, underlying SNARK constructions such as Groth16 [Gro16] and PLONK [GWC19], the KZG polynomial commitment scheme [KZG10], and BLS signatures [BGW+22]. While native pairing computation is well-studied and highly optimised, a different challenge arises when a pairing must be verified *inside another proving system*—that is, when the pairing computation itself becomes a statement to be proved in an arithmetised circuit.

This setting arises naturally in **proof recursion**, where one SNARK verifies another, and more broadly whenever a verifier is itself constrained: in resource-metered execution environments, in on-chain verification, and in any arithmetised computation system (R1CS [GGPR13], Plonkish [GWC19], AIR [GPR21]) where each $\mathbb{F}_{q^{12}}$ operation translates

into many constraints. In these contexts, the cost of expressing a pairing as a circuit dominates verification, and reducing it has been the subject of a substantial line of work.

1.1 Our Contribution

We present a public-coin interactive proof system for verifying pairing computations that consolidates verification into unified polynomial relationships under the Schwartz-Zippel lemma. By treating complete Miller loop iterations as single polynomial relationships rather than sequences of individual extension field operations [Fel23], our approach achieves:

- **39% reduction** in Plonkish constraints ($1,890,771 \rightarrow 1,158,194$ for BN254 Groth16 simulation)
- **60% reduction** in communication complexity ($739,198 \rightarrow 295,946$)
- **$\sim 42\%$ reduction** in compile-time memory and up to **52%** in solver memory

Our key insight is that instead of verifying individual field operations separately, we can verify entire Miller loop iterations as single polynomial identities. This consolidation dramatically reduces both the number of modular operations required and the communication overhead from intermediate commitments.

As a reusable artefact of independent interest, we also contribute a polynomial ring toolkit for gnark’s emulated arithmetic package. The toolkit instantiates the two-round interactive proof system generically: it is parameterised by an arbitrary modulus polynomial $P \in \mathbb{F}_q[x]$ and an emulated base field, so the same construction applies to pairing verification on any curve and, more broadly, to any computation that reduces to multiplications in $\mathbb{F}_q[x]/\langle P \rangle$. Multiplications submitted to the toolkit are verified lazily: the prover’s quotient and remainder are computed outside the circuit, and the resulting polynomial-identity check is deferred to circuit finalisation, where all checks under a common modulus are batched into a single Schwartz-Zippel test against two Fiat-Shamir challenges.

1.2 Related Work

The landscape of efficient pairing verification in constraint systems has evolved through three key contributions that our work directly builds upon.

El Housni’s **Pairings in Rank-1 Constraint Systems** [Hou23] established the theoretical framework for optimizing pairings in arithmetized environments, introducing efficient Miller loop formulas in affine coordinates and sparse line function representations. This work inverts the usual cost model: inversions become cheap (2C) while multiplications remain the dominant cost.

Feltroidprime [Fel23] introduced polynomial identity testing for $\mathbb{F}_{q^{12}}$ arithmetic via the Schwartz-Zippel lemma. Each multiplication $A \cdot B$ is expressed as $A(x) \cdot B(x) = Q(x) \cdot P_{12}(x) + R(x)$ and verified at a single random point, with the quotients Q batched via random linear combination. Our work applies the same polynomial framework but at a coarser granularity: entire Miller loop steps rather than individual multiplications, eliminating intermediate remainder commitments entirely.

Novakovic and Eagen’s **On Proving Pairings** [NE24] showed that final exponentiation can be replaced by a residue witness check, reducing final exponentiation cost by over 85% for BN254. We integrate their residue witness c directly into our polynomial step equations (Section 3.2.2), unifying the Miller loop and final exponentiation into the same polynomial verification framework.

These works provide the baseline for our construction. Our work introduces optimizations that further reduce constraint counts beyond the cumulative gains of these existing techniques.

2 Background

Algorithm 1 Optimal Ate pairing over Barreto-Naehrig curves [BGM⁺10]

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2$

Output: $a_{\text{opt}}(Q, P)$

```

1: Write  $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$  where  $s_i \in \{-1, 0, 1\}$  and  $s_L = 1$ 
2:  $T \leftarrow Q, f \leftarrow 1$ 
3: for  $i = L - 2$  to 0 do
4:    $f \leftarrow f^2 \cdot l_{T,T}(P)$ 
5:    $T \leftarrow [2]T$ 
6:   if  $s_i = 1$  then
7:      $f \leftarrow f \cdot l_{T,Q}(P)$ 
8:      $T \leftarrow T + Q$ 
9:   end if
10:  if  $s_i = -1$  then
11:     $f \leftarrow f \cdot l_{T,-Q}(P)$ 
12:     $T \leftarrow T - Q$ 
13:  end if
14: end for
15:  $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$ 
16:  $f \leftarrow f \cdot l_{T,Q_1}(P), T \leftarrow T + Q_1$ 
17:  $f \leftarrow f \cdot l_{T,-Q_2}(P), T \leftarrow T - Q_2$ 
18:  $f \leftarrow f^{(p^{12}-1)/r}$ 
19: return  $f$ 

```

2.1 Pairing Computation

An elliptic curve pairing is a non-degenerate bilinear map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$, where $\mathbb{G}_1$ and $\mathbb{G}_2$ are groups of points on elliptic curves over finite fields, and $\mathbb{G}_T$ is a multiplicative group in an extension field. For practical cryptographic applications, we require that this map be efficiently computable. We will focus on the optimal ate pairing [BGM⁺10] over Barreto-Naehrig (BN) curves, which is widely used due to its efficiency. We will specifically use the Ethereum’s BN254 curve, defined over a 254-bit prime field $\mathbb{F}_q$, with groups $\mathbb{G}_1, \mathbb{G}_2$ of order r and target group $\mathbb{G}_T$ contained in the 12th extension field $\mathbb{F}_{q^{12}}$.

The computation of an optimal ate pairing—the focus of our implementation—consists of two primary phases:

1. **Miller loop:** The core iterative computation that evaluates line functions at pairs of points from $\mathbb{G}_1$ and $\mathbb{G}_2$, accumulating intermediate results to produce a value in $\mathbb{F}_{q^{12}}$.
2. **Final Exponentiation:** This step raises the Miller loop output to the power $(q^{12} - 1)/r$, where r is the order of the groups, ensuring the result lies in the correct subgroup $\mathbb{G}_T$.

While both phases present optimization opportunities, the constraint-based verification of these computations introduces unique challenges and possibilities that differ significantly from native implementations.

2.2 Schwartz-Zippel lemma

The Schwartz-Zippel lemma (or more precisely, the Demillo-Lipton-Schwartz-Zippel lemma [DL78, Sch79, Zip79, Sch80]) provides a probabilistic method for testing polynomial identities. The lemma states that for a non-zero polynomial $P(x_1, x_2, \dots, x_n)$ of total degree d over a finite field $\mathbb{F}$, when evaluated at uniformly random field elements, the probability that P evaluates to zero is at most $d/|\mathbb{F}|$.

This probabilistic bound becomes negligible for cryptographically sized fields where $d \ll |\mathbb{F}|$, making it practical to verify polynomial identities by evaluation instead of coefficient-wise operations.

2.3 Computation Models

We implement and benchmark our toolkit in gnark [Cona], the industry-standard Go library for arithmetized circuit development, targeting both SCS (Sparse Constraint System, the native Plonkish backend) and R1CS constraint systems. We map each field multiplication to a constraint (denoted by **C**) to maintain compatibility with the R1CS model used in established literature. In R1CS [Gro16], prover complexity is dominated by multiplication gates while additions are free. To synchronize our theoretical analysis with the benchmarks provided by El Housni [Hou23], we define our cost metric based on field multiplications.

2.3.1 R1CS Cost Model

We measure computational cost in terms of *constraints* (denoted by **C**), representing the fundamental unit of work in Rank-1 constraint systems. One constraint corresponds to one multiplication gate.

Basic Field Operations in $\mathbb{F}_q$:

- **Addition/Subtraction:** Free
- **Multiplication:** One constraint (1C)
- **Division/Inversion:** One multiplication + one equality check

Commitment Overhead: Beyond constraint costs, we account for *calldata* or *commitment overhead*—the number of field elements the prover must commit to (via Fiat-Shamir hashing in non-interactive proofs). For blockchain deployments, this overhead directly impacts transaction costs and is often the dominant factor.

2.3.2 Cost Model Comparison

Operations that are traditionally expensive (inversions, divisions) become cheap in constraint systems when computed via witnesses. This inverts traditional optimization priorities: we can afford inversions to save multiplications, enabling affine coordinate systems and other techniques impractical in native implementations.

Recognizing and exploiting these unique computational properties is essential for unlocking hidden performance potential. Youssef El Housni’s pioneering work **Pairings in Rank-1 Constraint Systems** [Hou23] represents, to our knowledge, the first systematic exploration of these optimization opportunities, providing a breakthrough in the practicality of recursive proving by thoughtfully leveraging the distinctive cost model inherent to constraint-based systems.

3 Pairing Operations as Polynomials

Algorithm ?? gives the pairing computation combining El Housni's R1CS optimizations [Hou23] with Novakovic-Eagen's residue witness technique [NE24]. The 27th root of unity W scales the Miller loop output f to be a cubic residue (see [NE24] §4.3); if f is not already a cubic residue, multiplying by W^w , $w \in \{1, 2\}$ corrects this.

Our contribution is to express each step of this algorithm as a single multi-operand polynomial product verified via the Schwartz-Zippel lemma, rather than as a sequence of individual binary multiplications as in [Fel23].

Algorithm 2 Optimal Ate pairing with Residue witness check

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2, c \in \mathbb{F}_{q^{12}}, w \in \{0, 1, 2\}$
 Internal $W, W^2 \in \mathbb{F}_{q^{12}}$ ▷ W is 27th root of unity
Output: $a_{opt}(Q, P)$

- 1: Write $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$ where $s_i \in \{-1, 0, 1\}$ and $s_L = 1$
- 2: $c^{-1} \leftarrow 1/c$ ▷ Inverse Residue witness
- 3: $f \leftarrow c^{-1}$ ▷ Initialize with inverse residue witness
- 4: $T \leftarrow Q$ ▷ T_i for each pair for multi-pairing
 Every operation involving T, P or Q is repeated for T_i, P_i and Q_i for multi-pairing.
- 5: **for** $i = L - 2$ to 0 **do**
- 6: $f \leftarrow f^2$
- 7: $f \leftarrow f \cdot l_{T,T}(P)$ ▷ Repeat for T_i, P_i
- 8: $T \leftarrow [2]T$ ▷ Repeat for T_i
- 9: **if** $s_i = 1$ **then**
- 10: $f \leftarrow f \cdot c^{-1}$ ▷ Residue witness
- 11: $f \leftarrow f \cdot l_{T,Q}(P)$ ▷ Repeat for T_i, Q_i, P_i
- 12: $T \leftarrow T + Q$ ▷ Repeat for T_i, Q_i
- 13: **else if** $s_i = -1$ **then**
- 14: $f \leftarrow f \cdot c$ ▷ Residue witness
- 15: $f \leftarrow f \cdot l_{T,-Q}(P)$ ▷ Repeat for $T_i, -Q_i, P_i$
- 16: $T \leftarrow T - Q$ ▷ Repeat for $T_i, -Q_i$
- 17: **end if**
- 18: **end for**
- 19: $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$ ▷ Repeat for Q_{1i}, T_i
- 20: $f \leftarrow f \cdot l_{T,Q_1}(P)$ ▷ Repeat for Q_{1i}, T_i, P_i
- 21: $T \leftarrow T + Q_1$ ▷ Repeat for Q_{1i}, T_i
- 22: $f \leftarrow f \cdot l_{T,-Q_2(Q)}(P)$ ▷ Repeat for Q_i, T_i, P_i
- 23: **if** $w = 1$ **then**
- 24: $f \leftarrow f \cdot W$
- 25: **else if** $w = 2$ **then**
- 26: $f \leftarrow f \cdot W^2$
- 27: **end if**
- 28: $f \leftarrow f \cdot c^{-q} \cdot c^{q^2} \cdot c^{-q^3}$ ▷ Uses Frobenius maps, Negative exponents use c^{-1}
- 29: **return** f

3.1 Comparative Analysis

We compare the two schemes for a 3-pair Miller loop zero-bit step (the most common case in BN254's NAF[DHM04] representation of $s = 6t + 2$). Here $F \in \mathbb{F}_{q^{12}}$ is the accumulator,

163 $L_i \in \text{Sparse}_{01379}$ are the three doubling line functions, $R \in \mathbb{F}_{q^{12}}$ is the updated accumulator,
 164 and Q is the quotient polynomial. Sparse_{01379} denotes the subspace of $\mathbb{F}_{q^{12}}$ elements whose
 165 only nonzero coefficients are at degrees 0, 1, 3, 7, 9—the form that BN254 line functions
 166 naturally take [Hou23], reducing their evaluation cost from 12C to 4C. All compared
 167 schemes—[Hou23], [Fel23], and ours—exploit this sparseness of the Miller line functions;
 168 the corresponding savings are included in every constraint count below.

169 3.1.1 Previous scheme

170 Adapting [Fel23] to lines 6–7 of Algorithm ?? for a 3-pair Miller loop gives:

$$\begin{aligned}
 171 \quad & F(x) \cdot F(x) \stackrel{?}{=} Q_1(x) \cdot P_{12}(x) + T_1(x) \\
 172 \quad & T_1(x) \cdot L_1(x) \stackrel{?}{=} Q_2(x) \cdot P_{12}(x) + T_2(x) \\
 173 \quad & T_2(x) \cdot L_2(x) \stackrel{?}{=} Q_3(x) \cdot P_{12}(x) + T_3(x) \\
 174 \quad & T_3(x) \cdot L_3(x) \stackrel{?}{=} Q(x) \cdot P_{12}(x) + R(x)
 \end{aligned}$$

175 Evaluating F , T_1 – T_3 , R costs 12C each and L_1 – L_3 cost 4C each; all four remainder
 176 polynomials require commitment. **Total: 76C and 48 $\mathbb{F}_q$ commitments** (4×12).

177 3.1.2 Our scheme

178 Our scheme consolidates lines 6–7 into a single polynomial identity:

$$F(x) \cdot F(x) \cdot L_1(x) \cdot L_2(x) \cdot L_3(x) \stackrel{?}{=} Q(x) \cdot P_{12}(x) + R(x)$$

179 By eliminating intermediate T_* polynomials, only R requires commitment. **Total:**
 180 **41C and 12 $\mathbb{F}_q$ commitments** ($2 \times 12C + 3 \times 4C + 5C$).

181
 182 This results in **46% fewer constraints** and a **75% reduction in commitments**.
 183 When verifying ZK proofs on-chain, commitment savings are often more critical than
 184 constraint savings, due to the calldata costs associated with transmitting polynomial hints.

185 3.1.3 Performance Comparison

186 Table 1 compares the computational costs and commitment overhead between the schemes
 187 for a single 0-bit Miller loop operation with 3 pairs.

Table 1: 3-Pair Miller loop 0-bit constraints comparison

Scheme	Constraints	$\mathbb{F}_q$ Commitments	Improvement
[Hou23]	132C	—	Baseline
[Fel23]	76C*	48	42%
Our scheme	41C*	12	69%
[Hou23]: 1 square (36C) + 3 sparse mul ($3 \times 30C$) = 132C [Fel23]: 5 evals ($5 \times 12C$) + 3 line evals ($3 \times 4C$) + 4 muls ($4 \times 1C$) = 76C Our scheme: 2 evals ($2 \times 12C$) + 3 line evals ($3 \times 4C$) + 5C (muls + RLC) = 41C *Costs exclude Fiat-Shamir commitment overhead			

3.2 Operation Polynomials

Our implementation uses different polynomials for different bits in the Miller loop. Let us walk through a 3-pairing setup, the configuration used for Groth16 verification (Section ??). We will describe the polynomials for zero bits, non-zero bits and correction step.

3.2.1 Miller loop: Zero Bit Equation

As seen in 3.1 for zero bit operations (line 6 and 7 with three $l_{T,T}(P)$ lines in Algorithm ??), the verification equation is:

$$F^2(x) \cdot L_1(x) \cdot L_2(x) \cdot L_3(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (1)$$

Table 2: Zero bit operation polynomial components

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Miller loop accumulator	$11 \times 2 = 22$	12
$L_1, L_2, L_3 \in \text{Sparse}_{01379}$	The line functions	$9 \times 3 = 27$	$4 \times 3 = 12$
$R \in \mathbb{F}_{q^{12}}$	Remainder, The result	11	12
Total constraints	36C Evals, 4C Muls and 1C RLC		41
Q (<i>Amortized See 4.1</i>)	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	38	39

3.2.2 Miller loop: Non-Zero Bit Equation

For non-zero bit operations (lines 10, 11, 14, 15 with three pairs of lines in Algorithm ??), the Miller loop performs both doubling and addition steps. So for each pair we have two Sparse_{01379} lines. The verification equation becomes:

$$F^2(x) \cdot W(x) \cdot L_{d1}(x) \cdot L_{d2}(x) \cdot L_{d3}(x) \cdot L_{a1}(x) \cdot L_{a2}(x) \cdot L_{a3}(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (2)$$

Table 3: Non-zero bit operation polynomial components

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Miller loop accumulator	$11 \times 2 = 22$	12
$L_{d1}, L_{d2}, L_{d3} \in \text{Sparse}_{01379}$	Doubling line functions	$9 \times 3 = 27$	$4 \times 3 = 12$
$L_{a1}, L_{a2}, L_{a3} \in \text{Sparse}_{01379}$	Addition line functions	$9 \times 3 = 27$	$4 \times 3 = 12$
W or $W^{-1} \in \mathbb{F}_{q^{12}}$	Residue witness	11	12
$R \in \mathbb{F}_{q^{12}}$	Remainder, The result	11	12
Total constraints	60C Evals, 8C Muls and 1C RLC		69
Q (<i>Amortized See 4.1</i>)	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	76	77

The residue witness W encapsulates the equivalence class relationship established by Novakovic and Eagen’s final exponentiation elimination. The inverse of W is used for negative bits, this just changes the coefficients the equation remains the same.

3.2.3 Miller loop: Frobenius mapping Equation

The correction step finalizes the Miller loop by incorporating the Frobenius endomorphism corrections (lines 20 and 22 in Algorithm ??). Again, we have two Sparse_{01379} lines for each pair, for π_p and $-\pi_p^2$ mappings, without any squaring of the accumulator. The verification equation becomes:

$$F(x) \cdot L_{1\pi}(x) \cdot L_{2\pi}(x) \cdot L_{3\pi}(x) \cdot L_{1\pi^2}(x) \cdot L_{2\pi^2}(x) \cdot L_{3\pi^2}(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (3)$$

Table 4: Correction step polynomial components

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Miller loop accumulator	11	12
$L_{1\pi}, L_{2\pi}, L_{3\pi} \in \text{Sparse}_{01379}$	π_p correction lines	$9 \times 3 = 27$	$4 \times 3 = 12$
$L_{1\pi^2}, L_{2\pi^2}, L_{3\pi^2} \in \text{Sparse}_{01379}$	$-\pi_p^2$ correction lines	$9 \times 3 = 27$	$4 \times 3 = 12$
$R \in \mathbb{F}_{q^{12}}$	Remainder, The result	11	12
Total constraints	48C Evals, 6C Muls and 1C RLC		55
Q (<i>Amortized See 4.1</i>)	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	54	55

3.2.4 Final Exponentiation Equation

The final exponentiation check is handled as described in [NE24]. The equation handles the cubic scale factor multiplication, and **Frobenius component** ($q - q^2 + q^3$) of the final exponentiation (lines 24, 26 and 28 in Algorithm ??). The verification equation looks like:

$$F(x) \cdot c^{-q}(x) \cdot c^{q^2}(x) \cdot c^{-q^3}(x) \cdot W^w(x) = 1 + Q(x) \cdot P_{12}(x) \quad (4)$$

where $w \in \{0, 1, 2\}$ determines the cubic root of unity scaling factor, and the Frobenius mappings are computed using the residue witness c and its inverse and provided as input. Cubic root is sparse element with just 2 coefficients the degree ranging from 8 to 10 because of positioning of the coefficients. Our R is now 1 for a successful proof verification.

Table 5: Final exponentiation polynomial components

Polynomial	Description	Degree	Coefficients
$F \in \mathbb{F}_{q^{12}}$	Final Miller loop result	11	12
$c^{-q} \in \mathbb{F}_{q^{12}}$	Residue witness Frobenius $-q$	11	12
$c^{q^2} \in \mathbb{F}_{q^{12}}$	Residue witness Frobenius q^2	11	12
$c^{-q^3} \in \mathbb{F}_{q^{12}}$	Residue witness Frobenius $-q^3$	11	12
$W^w \in \mathbb{F}_{q^{12}}$	Cubic scale factor	10	2
$R = 1$	Unity (constant)	0	1
Total constraints	51C Evals, 4C Muls and 1C RLC		56
Q (<i>Amortized See 4.1</i>)	$\deg(\text{LHS}) - \deg(P_{12}) + 1$	43	44

With this we have all the polynomials we need for verifying entire pairing operation. Next we will look into high level overview of pairing verification via an interactive oracle proof.

3.3 Security

Each verification equation takes the form $P = \prod_i A_i(x) - Q(x) \cdot P_{12}(x) - R(x)$ where $\deg(P) < 2^8$.¹ By the Schwartz-Zippel lemma (Section 2.2):

Soundness: $\delta_s \leq d/|\mathbb{F}_q| < 2^8/2^{254} = 2^{-246}$ for BN254.

Completeness: An honest prover always produces $P(x) = 0$ identically, so the verifier always accepts. Euclidean division guarantees unique Q, R with $\deg(R) < \deg(P_{12}) = 12$.

¹For k -pair pairings, $\deg(P) \approx 88 + 18(k - 3)$, well below 2^8 for all practical k .

4 Full Pairing Proof with Quotient Batching

We now combine the polynomial equations from Section 3.2 into a complete two-round public-coin interactive proof, further reducing verification cost by batching all quotients into a single accumulated polynomial.

4.1 Batching Polynomial Identity Testing

All n polynomial equations from Section 3.2 are batched into a single verification. The prover commits to remainders $\{R_i\}_{i=0}^{n-1}$ and receives challenge z . Using z , the prover forms the accumulated quotient $Q_{\text{acc}} = \sum_{i=0}^{n-1} z^i \cdot Q_i$. The verifier then samples a second challenge point c and checks:

$$\sum_{i=0}^{n-1} z^i \cdot [A_i(c) \cdot B_i(c) \cdots X_i(c) - R_i(c)] = Q_{\text{acc}}(c) \cdot P_{12}(c) \quad (5)$$

This reduces n polynomial verifications (degrees 38–76, see Tables ??–??) to a single evaluation of a degree-76 polynomial.

4.2 Prover's Algorithms

The prover computes quotients Q_i and remainders R_i for each equation in Section 3.2. For each step, it calls POLYMULTIDIV (Algorithm 3), which multiplies the input polynomials and performs standard Euclidean division by the irreducible P_{12} . After receiving challenge z , the prover forms $Q_{\text{acc}} = \sum_i z^i Q_i$ and sends $\mathcal{R} = \{R_i\}$ and Q_{acc} to the verifier. The full prover procedure is given in Algorithm 4.

Algorithm 3 PolyMulDiv: Multiply polynomials and divide by P_{12}

Input: $P_1, \dots, P_n \in \mathbb{F}_q[x]$; Divisor $P_{12} \in \mathbb{F}_q[x]$

Output: Q, R s.t. $\prod P_i = Q \cdot P_{12} + R$, $\deg(R) < \deg(P_{12})$

- 1: $R \leftarrow \prod_{i=1}^n P_i$ $\triangleright$ Polynomial product
 - 2: Perform Euclidean division of R by P_{12} to get (Q, R)
 - 3: **return** (Q, R)
-

4.3 Verifier's Algorithms

The verifier receives $P \in \mathbb{G}_1$, $Q \in \mathbb{G}_2$, residue witness c [NE24], the W (27th root of unity), the remainder array $\mathcal{R}$, and Q_{acc} from the prover. Using challenge z from the first round and a freshly sampled x , it mirrors the prover's pairing loop while evaluating each polynomial at x via Algorithm 5, accumulating the RLC e_{acc} . The final check is Equation 5. The full verifier is given in Algorithm 6.

Algorithm 4 Pairing Prover: Preparing polynomials

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2, c \in \mathbb{F}_{q^{12}}, w \in \{0, 1, 2\}$
 Internal: $W, W^2 \in \mathbb{F}_{q^{12}}$ $\triangleright W$ is 27th root of unity

Output: Arrays $\mathcal{Q}, \mathcal{R}$ of quotients and remainders
 Write $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$ where $s_i \in \{-1, 0, 1\}$ and $s_L = 1$

- 1: $c^{-1} \leftarrow 1/c$ $\triangleright$ Inverse Residue witness
- 2: $f \leftarrow c^{-1}$ $\triangleright$ Initialize with inverse residue witness
- 3: $T \leftarrow Q$ $\triangleright T_i$ for each pair for multi-pairing

Every operation involving T, P or Q is repeated for T_i, P_i and Q_i for multi-pairing.

Miller loop bits

- 4: **for** $i = L - 2$ to 0 **do**
- 5: $L_{\text{dbl}} \leftarrow l_{T,T}(P), T \leftarrow [2]T$
- 6: **if** $s_i = 0$ **then**
- 7: $Q, R \leftarrow \text{PolyMulDiv}(f, f, L_{\text{dbl}})$ $\triangleright$ Algorithm 3, add L_{*i} for each P_i, Q_i
- 8: **else if** $s_i = 1$ **then**
- 9: $L_{\text{add}} \leftarrow l_{T,Q}(P), T \leftarrow T + Q$
- 10: $Q, R \leftarrow \text{PolyMulDiv}(f, f, c^{-1}, L_{\text{dbl}}, L_{\text{add}})$ $\triangleright$ Algorithm 3, add L_{*i} for each P_i, Q_i
- 11: **else if** $s_i = -1$ **then**
- 12: $L_{\text{add}} \leftarrow l_{T,-Q}(P), T \leftarrow T - Q$
- 13: $Q, R \leftarrow \text{PolyMulDiv}(f, f, c, L_{\text{dbl}}, L_{\text{add}})$ $\triangleright$ Algorithm 3, add L_{*i} for each P_i, Q_i
- 14: **end if**
- 15: $f \leftarrow R, \mathcal{Q}.\text{append}(Q), \mathcal{R}.\text{append}(R)$
- 16: **end for**

Miller loop Frobenius correction

- 17: $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$
- 18: $L_\pi \leftarrow l_{T,Q_1}(P)$
- 19: $T \leftarrow T + Q_1$
- 20: $L_{-\pi^2} \leftarrow l_{T,-Q_2}(P)$
- 21: $Q, R \leftarrow \text{PolyMulDiv}(f, L_\pi, L_{-\pi^2})$ $\triangleright$ Algorithm 3, add L_{*i} for each P_i, Q_i
- 22: $f \leftarrow R, \mathcal{Q}.\text{append}(Q), \mathcal{R}.\text{append}(R)$

Final exponentiation: Cubic scaling + Frobenius mappings

- 23: **if** $w = 0$ **then**
- 24: $Q, R \leftarrow \text{PolyMulDiv}(f, c^{-q}, c^{q^2}, c^{-q^3})$ $\triangleright$ Algorithm 3
- 25: **else if** $w = 1$ **then**
- 26: $Q, R \leftarrow \text{PolyMulDiv}(f, W, c^{-q}, c^{q^2}, c^{-q^3})$ $\triangleright$ Algorithm 3
- 27: **else if** $w = 2$ **then**
- 28: $Q, R \leftarrow \text{PolyMulDiv}(f, W^2, c^{-q}, c^{q^2}, c^{-q^3})$ $\triangleright$ Algorithm 3
- 29: **end if**
- 30: $\mathcal{Q}.\text{append}(Q), \mathcal{R}.\text{append}(R)$ $\triangleright$ Final $\mathcal{Q}$ and $\mathcal{R}$ accumulation, skip f
- 31: **return** $\mathcal{Q}, \mathcal{R}$

Algorithm 5 EvalPoly: Evaluate polynomial product minus remainder via Horner’s method

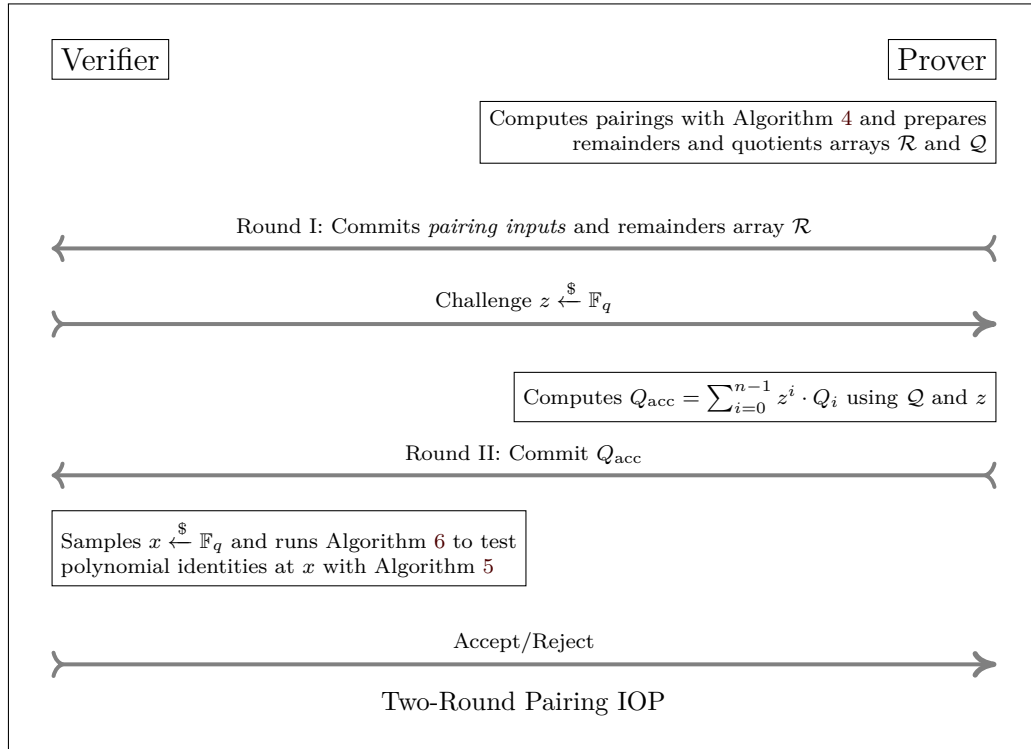
Input: Polynomials $P_1, \dots, P_n, R \in \mathbb{F}_q[x]$ of degree $\leq d$, evaluation point $z \in \mathbb{F}_q$

Output: $e \in \mathbb{F}_q$ where $e = \prod_{i=1}^n P_i(z) - R(z)$

- 1: **function** Eval ($P = (p_0, p_1, \dots, p_d), z$): **return** (*evaluation* $\in \mathbb{F}_q$)
 - 2: Uses Horner’s method, body not included for brevity
 - 3: $e \leftarrow \text{Eval}(P_1, z)$ ▷ Evaluate first polynomial
 - 4: **for** $i = 2$ to n **do**
 - 5: $e \leftarrow e \cdot \text{Eval}(P_i, z)$ ▷ Multiply in each subsequent evaluation
 - 6: **end for**
 - 7: $e \leftarrow e - \text{Eval}(R, z)$ ▷ Subtract remainder evaluation
 - 8: **return** e
-

4.4 Interactive Proof

252



4.5 Fiat-Shamir Transformation

255

256 Applying the Fiat-Shamir transformation [FS87] with random oracle H gives a non-
 257 interactive proof with knowledge soundness $\delta_s + 3(m^2 + 1)2^{-\lambda}$ [BSCS16], where $m = 2$
 258 (hash queries) and $\lambda = 254$ (BN254 hash output length).

259 **Gnark’s multi-commitment mechanism.** In gnark, each `api.Commit` call registers com-
 260 mitted variables in the constraint system; the two sequential calls in our protocol produce
 261 the two challenges z and x via gnark’s multi-round commitment scheme [BSB, BSB23].

Algorithm 6 Pairing Verifier: Evaluating remainders**Input:** $P \in \mathbb{G}_1, Q \in \mathbb{G}_2, c \in \mathbb{F}_{q^{12}}, w \in \{0, 1, 2\}, \mathcal{R}$ array of remainders, $Q_{\text{acc}} \in \mathbb{F}_q[x]$ Internal: 27th root of unity, $W, W^2 \in \mathbb{F}_{q^{12}}$, first round challenge, $z \xleftarrow{\$} \mathbb{F}_q$ **Output:** $\text{pairing}(P, Q) \stackrel{?}{=} 1$ Write $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$ where $s_i \in \{-1, 0, 1\}$ and $s_L = 1$

1: $c^{-1} \leftarrow 1/c$ ▷ Inverse Residue witness
 2: $f \leftarrow c^{-1}$ ▷ Initialize with inverse residue witness
 3: $T \leftarrow Q$ ▷ T_i for each pair for multi-pairing
 4: $x \xleftarrow{\$} \mathbb{F}_q$ ▷ Sample evaluation point from field
 5: $e_{\text{acc}} \leftarrow 0, a_{\text{rlc}} \leftarrow 1$ ▷ Evaluation accumulator and random linear combination multiplier
 Every operation involving T, P or Q is repeated for T_i, P_i and Q_i for multi-pairing.

Miller loop bits

6: **for** $i = L - 2$ to 0 **do**
 7: $L_{\text{dbl}} \leftarrow l_{T,T}(P), T \leftarrow [2]T$
 8: $R \leftarrow \mathcal{R}.\text{pop_front}()$ ▷ Fetch R from prover
 9: **if** $s_i = 0$ **then**
 10: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, f, L_{\text{dbl}})$ ▷ Algo 5, add L_{*i} for each P_i, Q_i
 11: **else if** $s_i = 1$ **then**
 12: $L_{\text{add}} \leftarrow l_{T,Q}(P), T \leftarrow T + Q$
 13: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, f, c^{-1}, L_{\text{dbl}}, L_{\text{add}})$ ▷ Algo 5, add L_{*i} for each P_i, Q_i
 14: **else if** $s_i = -1$ **then**
 15: $L_{\text{add}} \leftarrow l_{T,-Q}(P), T \leftarrow T - Q$
 16: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, f, c, L_{\text{dbl}}, L_{\text{add}})$ ▷ Algo 5, add L_{*i} for each P_i, Q_i
 17: **end if**
 18: $f \leftarrow R, a_{\text{rlc}} \leftarrow a_{\text{rlc}} \cdot z$ ▷ Update random linear combination factor
 19: **end for**

Miller loop Frobenius correction

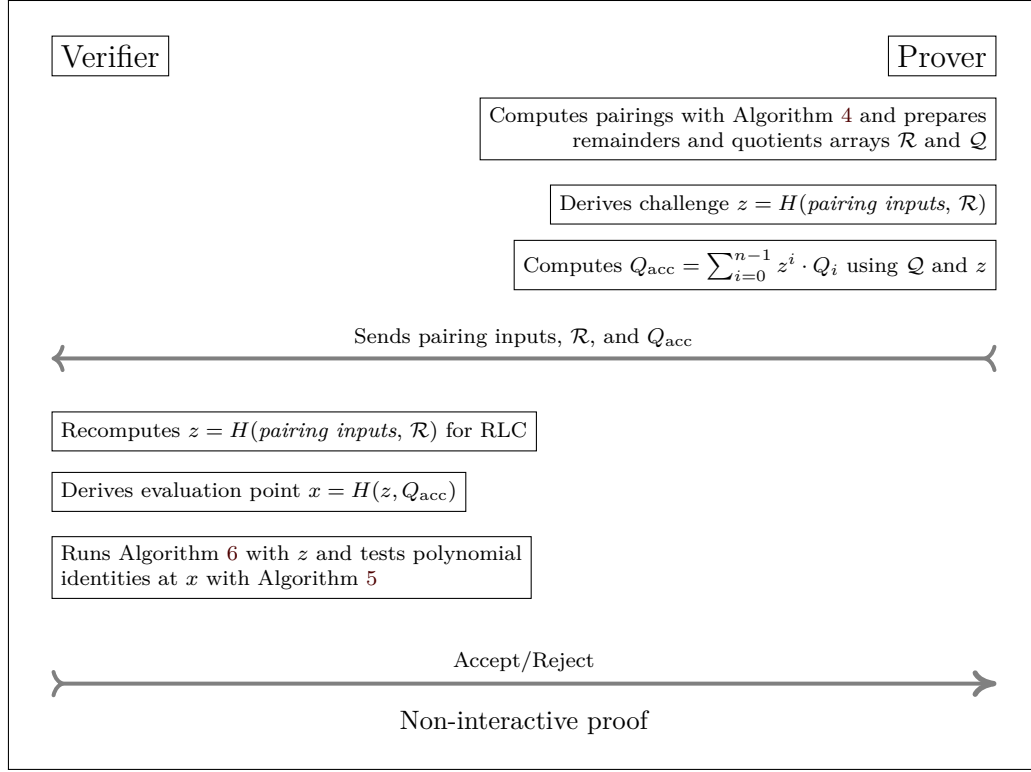
20: $R \leftarrow \mathcal{R}.\text{pop_front}()$ ▷ Fetch R from prover
 21: $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$
 22: $L_\pi \leftarrow l_{T,Q_1}(P)$
 23: $T \leftarrow T + Q_1$
 24: $L_{-\pi^2} \leftarrow l_{T,-Q_2}(P)$
 25: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, L_\pi, L_{-\pi^2})$ ▷ Algo 5, add L_{*i} for each P_i, Q_i
 26: $f \leftarrow R, a_{\text{rlc}} \leftarrow a_{\text{rlc}} \cdot z$ ▷ Update random linear combination factor

Final exponentiation: Cubic scaling + Frobenius mappings

27: $R \leftarrow \mathcal{R}.\text{pop_front}()$ ▷ Fetch R from prover
 28: **if** $w = 0$ **then**
 29: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, c^{-q}, c^{q^2}, c^{-q^3})$ ▷ Algo 5
 30: **else if** $w = 1$ **then**
 31: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, W, c^{-q}, c^{q^2}, c^{-q^3})$ ▷ Algo 5
 32: **else if** $w = 2$ **then**
 33: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, W^2, c^{-q}, c^{q^2}, c^{-q^3})$ ▷ Algo 5
 34: **end if**

35: **return** $Q_{\text{acc}}(x) \stackrel{?}{=} e_{\text{acc}} \wedge R \stackrel{?}{=} 1$ ▷ Polynomial identity and expected pairing output check

4.6 Non-Interactive Proof



Concrete commitment costs: In gnark, each call to `api.Commit` registers committed variables in the constraint system; the precise prover overhead depends on the backend (Groth16 or Plonk) [BSB, BSB23].

4.7 Security Analysis

Throughout this section, $z, x \in \mathbb{F}_q$ denote field element challenges (scalars), while polynomials are elements of $\mathbb{F}_q[X]$ with X as the formal indeterminate. In particular, z^j is a scalar coefficient, not a polynomial term.

4.7.1 Algebraic Intuition

Before the formal soundness proof it is instructive to consider why an incorrect remainder is difficult to hide. Suppose a malicious prover corrupts exactly one remainder: $R'_j = R_j + \Delta_j$ with $\Delta_j \in \mathbb{F}_q[X]$, $\Delta_j \neq 0$, $\deg(\Delta_j) \leq 11$. Since z is obtained *after* the commitment to $\mathcal{R}$, the prover cannot adjust any other R_k to compensate for Δ_j . To successfully deceive the verifier, the prover must find a valid forged quotient $Q'_{\text{acc}} \in \mathbb{F}_q[X]$ satisfying

$$\sum_{i=0}^{n-1} z^i (A_i - R'_i) = Q'_{\text{acc}} \cdot P_{12}$$

All $R'_i = R_i$ except for $R'_j = R_j + \Delta_j$. Substituting $R'_j = R_j + \Delta_j$ gives

$$\begin{aligned} Q'_{\text{acc}} \cdot P_{12} &= Q_{\text{acc}} \cdot P_{12} - z^j \Delta_j(x) \\ Q'_{\text{acc}} &= Q_{\text{acc}} - \frac{z^j \Delta_j(x)}{P_{12}(x)}. \end{aligned}$$

Here $z^j \in \mathbb{F}_q$ is a nonzero scalar (with overwhelming probability over the choice of z) and $\Delta_j(x) \in \mathbb{F}_q[X]$ is a nonzero polynomial of degree at most 11. Since P_{12} is irreducible of degree 12 and $\deg(z^j \Delta_j) \leq 11 < 12 = \deg(P_{12})$, the polynomial P_{12} does not divide $z^j \Delta_j$ in $\mathbb{F}_q[X]$; hence $z^j \Delta_j / P_{12}$ is not a polynomial. Consequently no valid $Q'_{\text{acc}} \in \mathbb{F}_q[X]$ can satisfy the equation for this single-corruption case.

The same degree argument extends immediately to an adversary who corrupts *multiple* remainders: for any scalar z , the weighted sum $\sum_i z^i \Delta_i(x)$ is still a polynomial in x of degree at most 11 (since each z^i is a scalar and each $\deg(\Delta_i) \leq 11$), so it cannot be divisible by P_{12} of degree 12 unless it is the zero polynomial, which requires every $\Delta_i = 0$. The formal soundness proof below re-derives this uniformly via Schwartz-Zippel.

4.7.2 Soundness

Theorem 1 (Soundness). *Let n be the number of polynomial equations batched via a random linear combination into the bivariate polynomial identity:*

$$P_{\text{final}}(z, x) = \sum_{i=0}^{n-1} z^i \cdot \left[\prod_j P_{i,j}(x) - R_i(x) \right] - Q_{\text{acc}}(x) \cdot P_{12}(x) = 0$$

Let the maximum degree in x of the polynomial products be $d_x < 2^8$. For a prover committing to incorrect remainders, the probability of successful forgery δ_s over randomly chosen $(z, x) \in \mathbb{F}_q^2$ is bounded by:

$$\delta_s \leq \frac{n - 1 + d_x}{|\mathbb{F}_q|}$$

Proof. If the prover commits to an incorrect remainder $R_i(x)$, the evaluation yields a non-zero error polynomial $P_{\text{err}}(z, x)$. The structure of $P_{\text{final}}(z, x)$ establishes the following degree bounds for $P_{\text{err}}(z, x)$:

- **Degree in z :** at most $n - 1$ (due to the z^i weighting).
- **Degree in x :** at most $d_x < 2^8$ (due to the polynomial products).

By the Schwartz-Zippel lemma, the probability that the non-zero polynomial P_{err} evaluates to zero is bounded by the total degree divided by the field size:

$$\Pr[P_{\text{err}}(z, x) = 0] \leq \frac{\deg_z(P_{\text{err}}) + \deg_x(P_{\text{err}})}{|\mathbb{F}_q|} \leq \frac{n - 1 + d_x}{|\mathbb{F}_q|}$$

□

Concrete parameters for BN254: With $n = 67$ equations and $d_x \leq 256$:

$$\delta_s \leq \frac{66 + 256}{2^{254}} < \frac{2^9}{2^{254}} = 2^{-245}.$$

Non-interactive proof via Fiat-Shamir : From Section 4.5, under the random oracle model the knowledge soundness error becomes [BSCS16]:

$$\delta'_s = \delta_s + 3(m^2 + 1) \cdot 2^{-\lambda}$$

where $m = 2$ (hash queries) and $\lambda = 254$ (hash output length), giving

$$\delta'_s \leq 2^{-245} + 15 \cdot 2^{-254} < 2^{-244}.$$

298 *Remark 1* (Correctness of polynomial evaluation). Theorem 1 assumes that the verifier
 299 evaluates all polynomials at the challenge point x without arithmetic error. The circuit
 300 implementation uses the deferred-reduction inner-product optimisation of Section 5.5
 301 (Section 5.5.1), which accumulates limb-wise products over $\mathbb{Z}$ before a single final reduction.
 302 Correctness of this optimisation requires that no intermediate limb value reaches the
 303 native field characteristic p . We verify this bound concretely for BN254 in Lemma 1
 304 (Section 5.5.1).

305 4.7.3 Completeness

306 For an honest prover executing Algorithm 4 with valid inputs:

1. **Polynomial construction:** At each operation, the prover computes following polynomials:

$$\prod_j P_{i,j} = Q_i \cdot P_{12} + R_i$$

2. **RLC formation:** Given challenge z , the prover forms:

$$Q_{\text{acc}} = \sum_{i=0}^{n-1} z^i \cdot Q_i$$

- 307 3. **Verifier's check:** The verifier evaluates at point x :

$$\begin{aligned} \text{LHS} &= \sum_{i=0}^{n-1} z^i \cdot \left[\prod_j P_{i,j}(x) - R_i(x) \right] \\ &= \sum_{i=0}^{n-1} z^i \cdot Q_i(x) \cdot P_{12}(x) \quad (\text{by construction}) \\ &= P_{12}(x) \cdot \sum_{i=0}^{n-1} z^i \cdot Q_i(x) \\ &= P_{12}(x) \cdot Q_{\text{acc}}(x) = \text{RHS} \end{aligned}$$

312 Since the equality holds for any evaluation point x , the protocol satisfies perfect
 313 completeness.

314 5 Polynomial Ring Toolkit

315 The protocol of Section 4 is packaged as a reusable toolkit in the `emulated` package of
 316 gnark [Conb], parameterised by any emulated field $\mathbb{F}_q$ (via `FieldParams` type T) and any
 317 modulus polynomial $P \in \mathbb{F}_q[x]$. The toolkit operates in a recursive proof setting where
 318 the prover's computations are performed outside the circuit via `hints` (Section 5.2). And
 319 verifier's checks are implemented as constraints inside the circuit.

320 Refer to our fork at <https://anonymous.4open.science/r/tiny-gnark-DDCD> for
 321 code.

322 5.1 Ring Representation

323 A ring element is essentially a coefficient slice `[]*Element[T]` in ascending degree order. A
 324 `nil` entry is treated as zero by `InnerProductNoReduce`—the term is skipped entirely—so
 325 sparse polynomials carry no wasted constraint variables.

326 The optional `modEvalFn` of type `EvalFnType[T] = ([*Element[T]]) → *Element[T]`
 327 allows handling small-integer coefficients efficiently by multiplying by the small positive
 328 integer first and negating afterwards, rather than multiplying by the full-width field
 329 representative $q - c$.

330 **Example: BN254 $\mathbb{F}_{q^{12}}$**

331 $P_{12}(x) = x^{12} - 18x^6 + 82$ has nine zero coefficients (stored as `nil`) and two small non-zero
 332 ones. The `modEvalFn` computes $P_{12}(\alpha) = \text{MulConst}(\alpha^0, 82) + \text{Neg}(\text{MulConst}(\alpha^6, 18)) + \alpha^{12}$,
 333 keeping intermediate values small instead of multiplying by $q - 18$.

334 5.2 Hint Functions

335 Hints allow defining computations outside of a circuit. A hint defines an annotated
 336 function in the circuit at compile time. Inputs and outputs are injected at the solving
 337 time. We define these two hints to implement the prover's steps in the two-round protocol
 338 of Section 4.4.

339 **polyRingMulHint** implements `POLYMULDIV` (Algorithm 3): it multiplies the input
 340 polynomials, divides by P to obtain (Q, R) , and serialises the result as `nbQTerms` | Q limbs |
 341 `nbRemTerms` | R limbs. Inputs are packed as `nbBits` | `nbLimbs` | `nbPoly` | q | inputs | P ,
 342 where each polynomial is prefixed by its term count.

343 **quotientsRLCHint** computes the accumulated quotient $Q_{\text{acc}}[j] = \sum_i z^i \cdot Q_i[j] \bmod q$
 344 coefficient-wise, corresponding to the prover's step in Algorithm 4.

345 5.3 API

346 The toolkit exposes three entry points. `NewPolyRingCheck` registers a ring group (mod-
 347 ulus P plus an optional small-coefficient evaluator) and returns a handle shared by every
 348 multiplication under that modulus; multiple groups with different moduli may coexist.
 349 `MulPolyRings` computes $R = (\prod_j A_j) \bmod P$ outside the circuit via `polyRingMulHint`,
 350 range-checks R 's limbs inside, and enqueues the check $(A_j; R; Q)$ in the group; for longer
 351 product chains a `PolyRingAccumulator` wrapper automatically triggers intermediate re-
 352 ductions when the accumulated degree exceeds a configurable bound. At circuit finalisation,
 353 `performDeferredRingChecks` implements Equation (5) via two sequential `api.Commit`
 354 calls [BSB] (producing challenges z and x) and one `AssertIsEqual` per group, so no
 355 in-circuit work beyond remainder range-checks is performed until this point. Full API
 356 signatures and a step-by-step code walkthrough are available in the repository.²

358 5.4 Usage

359 Core API

360 Register the ring group once at initialisation:

```
361 fp, _ := emulated.NewField[emulated.BN254Fp](api)
362 modPoly := fp.MakePoly([]any{82, 0, 0, 0, 0, 0, -18, 0, 0, 0, 0, 0, 1})
363 ring := fp.NewPolyRingCheck(modPoly, nil)
364 // Each Miller-loop step then calls:
365 rem, _ = f.MulPolyRings(ring, f, f, L1, L2, L3)
```

²<https://anonymous.4open.science/r/tiny-gnark-DDCD>

366 and `performDeferredRingChecks` is invoked once by the gnark backend at finalisation,
 367 running the full two-round protocol of Section 4.

368 Accumulate a product chain

```

369 acc := f.NewPolyRingAccumulator(group, targetDeg)
370 acc.Mul(A)
371 acc.Mul(B)
372 acc.Sqr()      // enqueues (A·B)2 without multiplying yet
373 rem := acc.Eval()
```

374 5.5 Implementation Details and Performance

375 5.5.1 Schoolbook vs Horner’s Method for Polynomial Evaluation

Horner’s method for evaluations across emulated elements causes a linear increase in the limbs count per coefficient. This necessitates reductions after each multiplication. Instead, we cache powers of the evaluation point x in $\mathcal{X} = (x^0, x^1, \dots, x^d)$ and evaluate each polynomial as an inner product of $\mathcal{X}$ and the coefficient array $\mathbf{F}$:

$$\mathcal{X} \cdot \mathbf{F} = \mathcal{X}_0 \cdot F_0 + \mathcal{X}_1 \cdot F_1 + \dots + \mathcal{X}_d \cdot F_d$$

376 In the emulated field context, EVAL in Algorithm 5 is replaced by this inner product
 377 with $\mathcal{X}$. This allows us to use the inner product optimization described below.

378 5.5.2 Native field operations for inner products

379 Modular reductions can be deferred or entirely elided for operations expressible as inner
 380 products, such as polynomial evaluations and random linear combinations. This optimiza-
 381 tion leverages the capacity headroom provided by representing small emulated limbs (e.g.,
 382 64-bit limbs) within a significantly larger native field element.

383 **Deferred Reduction:** Intermediate scalars are permitted to grow over the integers $\mathbb{Z}$
 384 during limb-wise multiplications and linear accumulations.

385 **Preservation of Identity:** Algebraic identities are strictly preserved provided the
 386 maximum accumulated limb magnitude remains bounded below the native characteristic
 387 p .

388 **Handling Asymmetry:** In circuits with asymmetric multiplicative depths, operands
 389 may accumulate as disparate polynomial multiples of the emulated characteristic p_{emul} .
 390 Consequently, a terminal single-element reduction suffices for equivalence testing, bypassing
 391 the constraint overhead of intermediate normalization.

392 **Lemma 1** (Overflow safety for deferred reduction). *Let $\mathbb{F}_q$ be emulated with limbs of width*
 393 *b bits inside a native field $\mathbb{F}_p$. Consider evaluating a degree- d polynomial via an inner*
 394 *product over the cached power vector $\mathcal{X}$. Since limbs are accumulated independently, each*
 395 *result-limb is a sum of $d + 1$ products of two b -bit values, so its magnitude is bounded by*
 396 *$\beta = 2^{2b + \lceil \log_2(d+1) \rceil}$.*

397 *The deferred single-element reduction is safe whenever $\beta < \log_2 p$.*

398 **Corollary 1** (BN254 pairing instantiation). *For BN254 pairing verification: $b = 64$,*
 399 *$d \leq 256$, and $p \approx 2^{254}$. Then*

$$400 \quad \beta = 2 \cdot 64 + \lceil \log_2 257 \rceil = 128 + 9 = 137 < 254,$$

401 *so the deferred reduction is safe.*

Table 6: Performance improvements in Garaga: Cairo implementation on Starknet

MULMOD		ADDMOD		COMMITTS		VM STEPS	
Before	After	Before	After	Before	After	Before	After
BN254: 3-Pairing with Miller loop and final exponentiation							
19,142	12,656	21,686	15,142	5,689	2,807	338,678	212,902
33.9%		30.2%		50.7%		37.1%	
BLS12-381: 3-Pairing with Miller loop and final exponentiation							
16,484	11,287	20,669	15,420	5,421	3,167	325,757	219,155
31.5%		25.4%		41.6%		32.7%	
MULMOD/ADDMOD: Field operations in $\mathbb{F}_p$ via built-in function calls							
COMMITTS: Elements committed for Fiat-Shamir, see § ??							
VM STEPS: CairoVM Steps [Sta22] (CPU Cycles)							

6 Conclusion

We have shown that choosing the right granularity for polynomial identity testing—entire Miller loop steps rather than individual field multiplications—yields substantial and measurable improvements in arithmetized pairing verification. The key technique, collapsing a composite multi-operand product into a single Euclidean division check with batched quotient accumulation, is simple to implement and broadly applicable to other multi-step cryptographic computations in arithmetized environments.

6.1 Applications and Impact

Our approach delivers maximum impact in resource-constrained execution environments where custom auxiliary data can be provided for cheaper verification. **Recursive proof systems** benefit the most, greatly reducing outer circuit complexity through fewer commitments and operations. **ZKVMs** (RISC Zero, SP1, Jolt) benefit naturally as they express computation as constraint systems compatible with our polynomial framework. **Calldata-constrained blockchains** (Ethereum L2s, BitVM) also benefit, as fewer commitments mean smaller proof transcripts and associated calldata costs.

This is already implemented in-production in Garaga [FE] for CairoVM and is currently the standard for all pairing based ZK proofs on Starknet. The polynomial ring toolkit (Section 5) in gnark, submitted as a part of this work, will directly impact the `evmprecompiles/ecpair` precompile via `AssertMillerLoopAndFinalExpIsOne`.

6.2 Implementation and Performance

The Garaga implementation [FE] in Cairo/Starknet (2024) sufficiently reduced the commitment overhead and modulo operations (Table 2) to fit Groth16 proof verification within Starknet’s transaction limits for the first time. However, Cairo [Sta22] is a specialized environment with its own cost metrics, making direct comparison with mainstream ZK frameworks difficult. This work situates the same technique within gnark—the standard framework for field emulation in arithmetized circuits—providing a systematic benchmark against an industry baseline.

In Table 2, the *Before* column is Garaga verifying each extension field multiplication individually via the polynomial identity testing of [Fel23]; the *After* column is Garaga using our consolidated relations. MULMOD/ADDMOD count modular field operations, COMMITTS count Poseidon [?] input commitments for Fiat-Shamir, and VM STEPS count CairoVM CPU cycles—together a low-level estimate of computational complexity inside the virtual machine.

Our polynomial ring toolkit implementation in gnark is benchmarked in Table 4, measured via a 3-pairing product check (see below) on BN254 and BLS12-381. We compare our results against a **gnark baseline** using the upstream **emulated** package [Cona], which represents the state-of-the-art for emulated pairings and includes the optimizations of [Hou23] and [NE24].

The **This Work** numbers were obtained by compiling the exact same benchmark against our fork, replacing the baseline’s extension field multiplications with our new toolkit. No changes were made to the circuit definition or test harness. Both implementations exploit the sparseness of the Miller line functions (Section 3.1). The excess memory savings come from the commitment and deferred-check pipeline: fewer deferred checks reduce both committed values and hint data payloads.

Benchmark circuit. Both implementations are evaluated against the same gnark circuit: a 3-pairing multi-pairing check $\prod_{i=1}^3 e(P_i, Q_i) = 1$ via **PairingCheck**. This corresponds to the pairing workload of a Groth16 verification, which reduces to a product-of-pairings check over three variable pairs once the constant $e(\alpha, \beta)$ term from the trusted setup is folded in. The same 3-pairing configuration is used in the CairoVM benchmarks of Table 2, allowing a like-for-like comparison across environments. Test circuit can be found at `sw_bn254` package in `g16_simulation_test.go` file in our implementation repo - <https://anonymous.4open.science/r/tiny-gnark-DDCD/>.

6.3 Further Work

Several natural extensions of this work remain open. First, the framework can be readily extended to additional curves beyond BN254 and BLS12-381, notably **BW6-761** for recursive proof chain verification on Ethereum. A further direction is exploring the technique for native field pairings, such as BLS12-377 inside BW6-761 [Hou23], where the trade-off between batching savings and commitment costs warrants dedicated analysis.

Finally, the underlying technique generalizes beyond pairings to any setting involving repeated operations over the same polynomial rings. Another interesting instance is lattice-based cryptography in ZK-SNARKs: schemes such as ML-KEM and ML-DSA operate in cyclotomic rings over $\mathbb{Z}_q[x]/(x^n + 1)$. Though the small moduli typical of lattice based schemes don’t require active reductions, the polynomial identity testing would help in two significant ways. One, skip the need to do the in-circuit NTT transforms, and two, provide reduced coefficients ready for efficient packing for public inputs.

Exploring the cost-benefits of this approach in lattice-based ZK-SNARKs is another promising direction for future work.

References

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Table 7: Performance comparison: gnark implementation, 3-pairing product check on BN254 and BLS12–381

Metric	gnark Baseline	This Work	Reduction
BN254 – SCS (Plonkish) Constraint System			
Nb Constraints	1,592,440	1,035,550	35.0%
Nb $\mathbb{F}_q$ Committed	571,125	269,616	52.8%
Compile Time	615.04 ms	347.13 ms	43.6%
Compile Memory	1.33 GB	0.82 GB	38.3%
Solve Time	167.57 ms	97.30 ms	41.9%
Solve Memory	348.73 MB	181.24 MB	48.0%
BN254 – R1CS Constraint System			
Nb Constraints	491,447	316,167	35.7%
Nb $\mathbb{F}_q$ Committed	571,125	269,616	52.8%
Compile Time	617.63 ms	367.25 ms	40.5%
Compile Memory	1.60 GB	0.98 GB	38.8%
Solve Time	175.42 ms	106.36 ms	39.4%
Solve Memory	173.26 MB	112.72 MB	34.9%
BLS12–381 – SCS (Plonkish) Constraint System			
Nb Constraints	1,784,247	1,191,105	33.2%
Nb $\mathbb{F}_q$ Committed	660,160	300,104	54.5%
Compile Time	650.02 ms	410.48 ms	36.9%
Compile Memory	1.48 GB	0.97 GB	34.5%
Solve Time	180.05 ms	111.60 ms	38.0%
Solve Memory	362.14 MB	284.30 MB	21.5%
BLS12–381 – R1CS Constraint System			
Nb Constraints	545,510	368,434	32.5%
Nb $\mathbb{F}_q$ Committed	660,160	300,104	54.5%
Compile Time	686.95 ms	424.01 ms	38.3%
Compile Memory	1.74 GB	1.13 GB	35.1%
Solve Time	182.72 ms	121.15 ms	33.7%
Solve Memory	231.77 MB	118.61 MB	48.8%

Hardware: Apple M3 Pro (goarch:arm64), 36gb, macOS Tahoe 26.5.2

$\mathbb{F}_q$ Committed: Elements committed via `api.Commit` calls for Fiat-Shamir, see § ??.

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